Frame-wise Piano Note Classification using Deep Learning

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Abstract

Automatic music transcription (AMT) remains a challenging task, particularly when detecting multiple simultaneous notes in polyphonic piano recordings. This project tackles frame-wise multi-label classification of active piano notes (MIDI 21–108) using the MusicNet dataset, focusing on a subset of 39 solo piano Bach pieces. Two temporal deep learning architectures—Bidirectional GRUs and Transformer Encoders—are evaluated on Constant-Q Transform (CQT) and Wav2Vec2 representations, with a CNN trained on Mel spectrograms serving as a baseline. All models are implemented from scratch without relying on pre-trained components. The experiments highlight the effectiveness of Transformers with CQT inputs for capturing long-range dependencies, achieving the highest Active F1 score. These results suggest that lightweight Transformer architectures combined with pitch-aligned input features are a promising direction for efficient and accurate AMT systems, particularly under computational constraints. The study concludes with practical recommendations for future improvements, including domain-specific fine-tuning of embeddings, hybrid feature representations, and lightweight data augmentation strategies to address class imbalance and rare-note detection.

1 Introduction

Automatic Music Transcription (AMT) seeks to convert audio recordings into symbolic formats such as musical notes or sheet music. In the case of polyphonic piano music, this involves detecting multiple notes sounding simultaneously, a task that remains challenging due to harmonic overlap and temporal dependencies. Accurate transcription systems have widespread applications in music education, digital archiving, and interactive performance tools.

This study addresses the problem of frame-wise multi-label classification of active piano notes (MIDI 21–108) using deep learning models. Lightweight temporal architectures—Bidirectional GRUs and Transformers—are evaluated and compared against a CNN baseline trained on Mel-spectrograms. All models, including the Transformer, GRU, and CNN, are implemented from scratch without the use of pre-trained components, ensuring full control over architectural and training parameters. Two input representations are considered: the pitch-aligned Constant-Q Transform (CQT) and self-supervised Wav2Vec2 embeddings. A stylistically consistent subset of the MusicNet dataset, limited to solo piano pieces by Bach, is used to control for musical variability and to enable focused evaluation under harmonically uniform conditions.

Experimental results indicate that Transformers using CQT inputs achieve the highest transcription performance, while models trained on Wav2Vec2 embeddings underperform in this domain. The study further investigates class imbalance and training dynamics, highlighting key trade-offs between representation type, model complexity, and predictive accuracy.

Preprint. Under review.

The remainder of this report is structured as follows: Section 2 reviews related work in AMT and deep learning-based transcription. Section 3 describes the dataset, preprocessing pipeline, model architectures, and training setup. Section 4 presents results and evaluation. Section 5 concludes with a summary of findings and outlines directions for future work.

2 Related Work

Automatic Music Transcription (AMT) is a longstanding challenge in Music Information Retrieval (MIR), focused on converting audio signals into structured symbolic representations such as sheet music Jamshidi et al. [2024]. Traditional approaches, including signal processing methods, probabilistic models, and Non-Negative Matrix Factorization (NMF), have attempted to separate complex polyphonic audio into individual note components. However, these methods often struggle with overlapping harmonics and dense musical textures.

Recent advances have been driven by deep learning, particularly convolutional and recurrent neural networks. Architectures such as Onsets and Frames Hawthorne et al. [2018] demonstrated strong performance by explicitly modeling both note onsets and frame-wise pitch activations. Transformer-based models have also been explored, leveraging attention mechanisms to better capture long-range temporal dependencies Jamshidi et al. [2024], Ou et al. [2022]. In particular, Ou et al. [2022] showed that incorporating Transformers can improve tasks such as velocity estimation in piano transcription, leading to gains at both frame- and note-level evaluations.

Most modern AMT systems operate on time-frequency representations such as Mel-spectrograms or Constant-Q Transform (CQT) spectrograms Jamshidi et al. [2024]. While Mel-spectrograms provide perceptually motivated frequency scaling, CQT offers improved resolution at lower frequencies, aligning more naturally with musical pitch structures Brown [1991]. However, the relative advantages of different input representations—especially for lightweight sequence models—remain underexplored. Additionally, although self-supervised embeddings such as Wav2Vec2 Baevski et al. [2020] have achieved success in speech processing, their application to AMT tasks has been limited.

In this project, we investigate the impact of input representations (CQT vs. Wav2Vec2) and lightweight sequence models (Transformer, GRU) on frame-wise piano note classification. While prior work has demonstrated strong results with time-frequency representations and large models, the effectiveness of lightweight architectures combined with emerging self-supervised embeddings remains underexplored for AMT tasks.

3 Experimental Setup

3.1 Dataset and Preprocessing

A subset of the MusicNet dataset was used, focusing exclusively on solo piano pieces composed by Bach. Bach was selected due to his highly structured and harmonically consistent compositions, which provide a more predictable and dense distribution of piano notes compared to more diverse composers like Beethoven.

Each audio file was processed into two types of input representations:

- **Constant-Q Transform (CQT):** Produces a pitch-aligned time-frequency representation, sized [frames, 88].
- **Wav2Vec2 Embeddings:** High-level learned features extracted per frame, sized [frames, 768].

Ground truth labels were generated as frame-wise multi-hot vectors, corresponding to active piano notes (MIDI notes 21–108).

Data was split into **Train**, **Validation**, and **Test** sets based on full pieces (not frames) to prevent leakage through repeated passages. A pre-defined, seeded split ensures reproducibility across runs. To guarantee that validation and test sets contained only notes seen during training, note coverage across splits was verified. Notes not appearing in the training set (approximately 39 out of 88) were retained in the output dimension to maintain MIDI alignment and potential generalisation. To preserve temporal structure, no data shuffling was applied during preprocessing or batching. Since note activation depends on the local musical context, maintaining frame order was essential for learning meaningful temporal dependencies.

3.2 Class Imbalance Investigation

To assess class imbalance, the activation frequency of each note across Train, Validation, and Test splits was analysed.

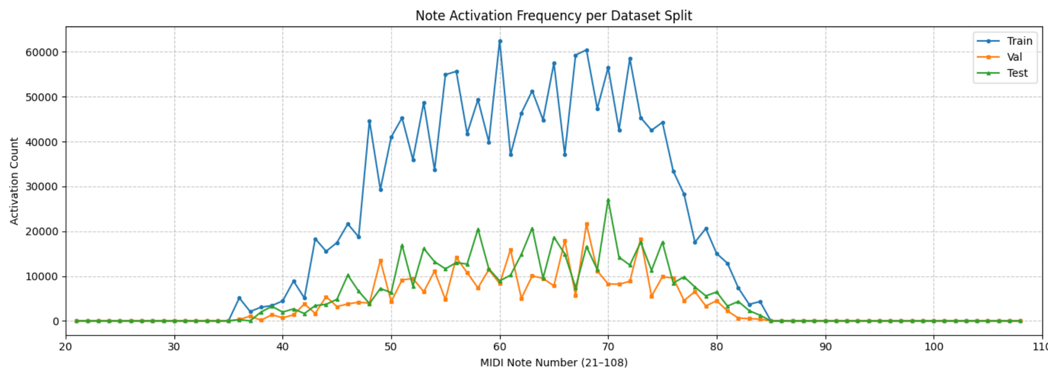


Figure 1: Label activation frequency across notes (MIDI 21–108) showing class imbalance in the dataset.

As shown in **Figure 1**, the distribution is heavily skewed towards mid-range notes (MIDI 50–70), while extreme low and high notes (MIDI 21–35 and 85–108) were completely inactive. Among notes with at least one activation, the imbalance ratio was **30.02**, indicating that the most frequent note appeared approximately 30 times more often than the least frequent activated note.

Although class imbalance corrections were explored after hyperparamter tuning, using a class-weighted loss (`pos_weight`), the improvements in Active F1 score were modest (approximately 1%). As the training, validation, and test distributions were consistent and reflective of real-world musical patterns, and the improvments are barely existant, the final models were trained **without applying pos_weight** to maintain natural prediction behavior.

3.3 Model Architectures

As a baseline, a simple CNN architecture operating on Mel-Spectrograms, the traditional input for audio tasks, was implemented from scratch. The architecture consisted of two convolutional blocks with max-pooling layers, a structure commonly used for early-stage audio classification tasks. This baseline ensured interpretability and provided a straightforward performance reference for more complex models. (See **Figure 2a**)

Building upon this baseline, more sophisticated architectures were explored to better capture the temporal and contextual structure of the audio data Two model families were implemented from scratch and evaluated across two input types (CQT and Wav2Vec2):

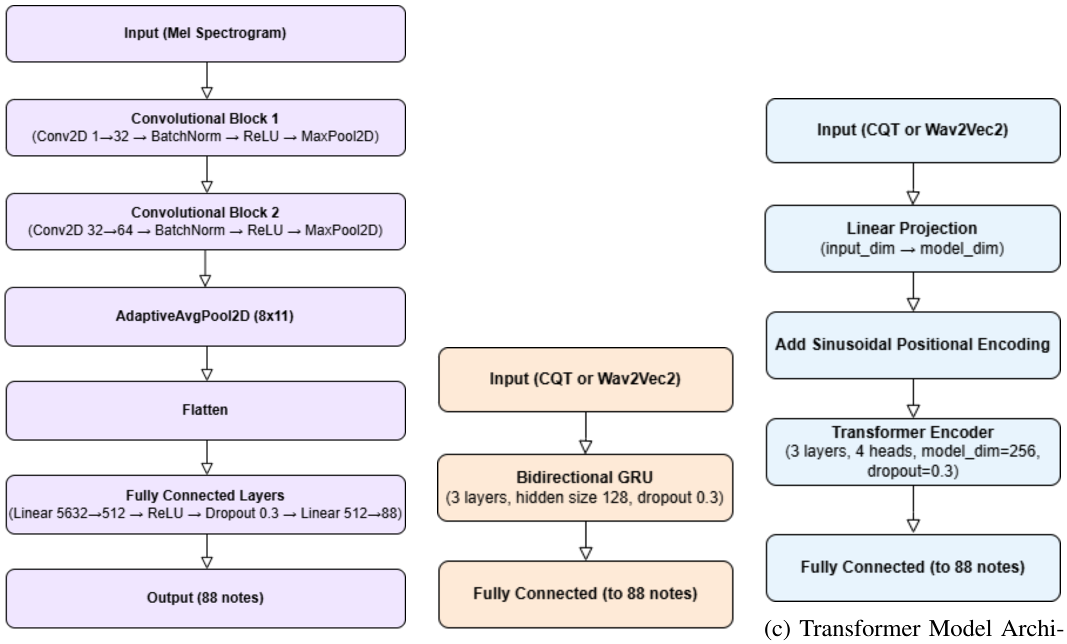
- **Transformer:** Lightweight version of the self-attention model.
- **GRU:** Sequential model leveraging recurrent dynamics.

Each model's input dimensionality matched the feature representation: 88 (CQT) or 768 (Wav2Vec2) for GRU and Transformer models.

The GRU model is a two-layer bidirectional recurrent network, capturing both past and future context to improve note prediction in temporal sequences. Forward and backward outputs are concatenated into a 256-dimensional representation per time step, followed by dropout to prevent overfitting. A final linear layer maps to 88 note activation scores. This architecture balances temporal modeling and computational efficiency.

The Transformer uses a lightweight encoder with stacked multi-head self-attention and feedforward layers. Positional encodings preserve temporal order in the input. Its ability to model frame-to-frame dependencies without recurrence makes it ideal for frame-wise note classification.

Architectural diagrams for each model are shown in Figures 2a, 2b, and 2c. The GRU and Transformer architectures presented below correspond to the final selected configurations following hyperparameter and model architecture tuning, while the CNN baseline architecture remained fixed.



(a) CNN Baseline Architecture

(b) GRU Model Architecture

(c) Transformer Model Architecture

Figure 2: Architectures of the CNN, GRU, and Transformer models used for frame-wise note classification.

3.4 Training Setup and Calibration

Training utilised **early stopping** based on validation Active F1 score. An initial experiment trained 3 representative models for 300-500 epochs, observing convergence general within 400 epochs. Therefore for the hyperparameter and architectural tuning setup, a max of 500 Epoches are chosen, with early stopping enables for 15 epoches of no improvemnet, with a condiation of at least 0.2 F1 achieved. For the final model evaluations, each model was trained for the full 500 epoches without any early stopping, to ensure there wasnt some unusual convergence behavior missed. Once it was confirmed there wasn't, each tuned model was trained using the usual early stopping mechanism, to prevent overfitting in the model. (See **Figure 3a** and **Figure 3b**)

After training, threshold sweeping (0.1–0.9) was used to determine the optimal sigmoid output threshold for each model based on validation Active F1. (See **Figure 4**)

3.5 Hyperparameter Tuning

A two-phase tuning strategy was adopted:

- **Phase 1 (Architecture & Hyperparameter Tuning):** Grid search over model-specific parameters: hidden sizes, number of layers, dropout rates, attention heads (Transformer), and learning rates. A sigmoid threshold of 0.3 was used across all models at this stage, at through experimentation it seemed to generally produce the best results per model.
- **Phase 2 (Calibration Tuning):** Fine-tuning sigmoid threshold and experimenting with `pos_weight` settings. However, this was only done on the final selected models, after hyperparameter tuning was complete.

As it was essential for each model to converge during training to get a fair comparison, early stopping was used to stop the model at convergence.

Selected hyperparameter configurations and their best validation F1 scores are summarized in **Table 1**.

Table 1: Selected Hyperparameter and Architecture Configurations given Best Validation F1 Scores

Input	Model	Hidden Size	Layers	Heads	Dropout	LR	Val F1
CQT	GRU	256	3	N/A	0.3	5e-4	0.672
CQT	Transformer	256	3	4	0.3	5e-4	0.727
Wav2Vec2	GRU	256	3	N/A	0.3	5e-4	0.374
Wav2Vec2	Transformer	256	3	4	0.3	5e-4	0.409

Interestingly, the best validation scores across all models corresponded to the highest values in the hyperparameter search space for key parameters such as hidden size (256), number of layers (3), attention heads (4), and dropout (0.3). This consistent trend suggests that the models benefited from increased capacity and regularisation, and that further performance gains might be achievable with even larger or deeper architectures. However, given computational constraints, the search space was capped conservatively—indicating the chosen values may not yet represent the true optima.

3.6 Training Speed Optimization

Despite the availability of powerful A100 GPUs, training remained relatively slow due to the large scale of the input data and the sequential nature of the models. For the CQT representation, the total number of frames was approximately **278,976**, each with 88 input dimensions. The Wav2Vec2 input had **161,913** frames per model, each of dimension 768. Although reducing the frame rate was considered, experiments and intuition suggested that high temporal resolution was essential for accurate note classification. Transitions between notes often occur within a very short time span, requiring fine-grained frame-wise modeling. The frame rate for CQT was around **86 FPS**, while Wav2Vec embeddings were extracted at a fixed **50 FPS**. This difference in temporal granularity may have contributed to CQT's superior performance, as the larger number of frames increased both model capacity and dataset richness.

To improve training efficiency and maximize GPU utilisation, the following strategies were applied:

- **Mixed Precision Training:** Automatic mixed precision (AMP) was used via PyTorch's `autocast` and `GradScaler`, reducing memory usage and improving throughput.
- **Large Batch Sizes:** Batch size was manually tuned to the maximum that could fit in GPU memory for each model/input type combination, ensuring stable training while maximizing parallelization.
- **Parallel Data Loading:** The DataLoaders were configured with 4 workers and pinned memory to reduce CPU-GPU transfer overhead.

These techniques led to approximately **25–35% reduction in training time** compared to a naive setup using full precision, small batch sizes, and single-threaded data loading.

3.7 Reproducibility

All experiments set random seeds (`torch.manual_seed(2025)`) for reproducibility. Preprocessing, splitting, and data transformations are fully scripted to enable re-running the pipeline end-to-end with consistent results.

4 Results

Active F1 is particularly well-suited to this task due to the strong class imbalance in note activations—most piano notes remain inactive across frames, while a small subset occurs frequently. Traditional accuracy would be inflated by this majority class, obscuring poor performance on active note prediction. In contrast, Active F1 directly evaluates the model's ability to detect note labels by balancing precision (avoiding false positives) and recall (capturing true activations), making it the most informative metric for transcription performance. It is widely used in automatic music transcription deep learning tasks for its effectiveness in multi-label, imbalanced settings.

4.1 Model Evaluation

A CNN baseline trained on Mel-spectrograms achieved a test F1 score of **0.63**, serving as a performance anchor for evaluating sequence models. **Table 2** reports full metrics.

Table 2: CNN Test Results (F1, Precision, Recall)

Metric	F1 Score	Precision	Recall
CNN (MelSpectrogram)	0.63	0.61	0.70

Table 3 summarizes the test performance across GRU and Transformer models trained on CQT and Wav2Vec2 inputs.

Table 3: GRU and Transformer Test Results Across Inputs

Model-Input	F1 Score	Precision	Recall
GRU (CQT)	0.63	0.66	0.62
GRU (Wav2Vec2)	0.29	0.30	0.31
Transformer (CQT)	**0.68**	0.72	0.67
Transformer (Wav2Vec2)	0.31	0.34	0.34

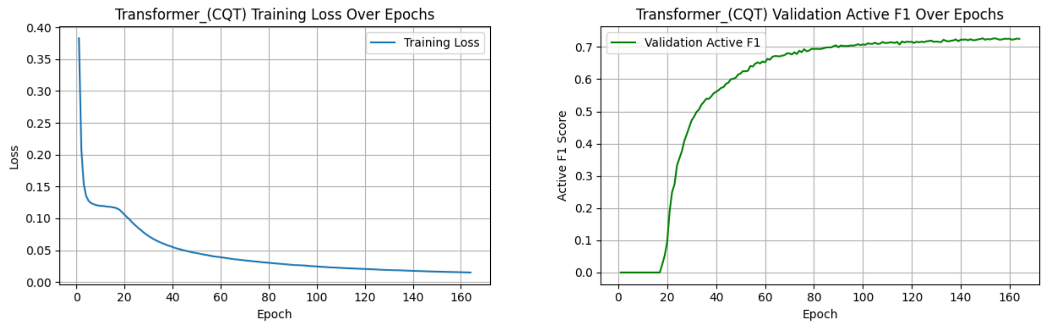
Key Findings:

- The best-performing model was the **Transformer with CQT input**, achieving an F1 score of 0.68. This represents a significant improvement over the CNN baseline (by 8%) and the GRU (by 5%), demonstrating the Transformer's superior ability to model long-range temporal dependencies in polyphonic audio.
- Models trained on **CQT consistently outperformed those using Wav2Vec2**, highlighting the value of pitch-aligned, high-resolution time-frequency representations for frame-wise note detection. CQT's logarithmic frequency spacing aligns well with the musical scale, likely making it easier for models to learn pitch-specific patterns.
- The **GRU and Transformer models both struggled with Wav2Vec2 inputs**, achieving F1 scores below 0.35. This suggests that Wav2Vec2, while powerful in speech tasks, may encode information less relevant to fine-grained note activation and pitch resolution. Its fixed 50 FPS frame rate and speech-centric pretraining likely introduce a mismatch with the dense, transient nature of piano music.

State-of-the-art systems such as Onsets and Frames Hawthorne et al. [2018] report note-level F1 scores exceeding 0.82. However, such models use onset supervision, larger architectures, and richer datasets. In contrast, this project focused on lightweight, frame-wise transcription with minimal preprocessing, making the CNN baseline a more appropriate point of comparison, which it exceeds by 8%.

Training Behavior & Threshold Tuning Impact

Training and validation losses (Figure 3a) showed smooth convergence for all models. Validation F1 peaked between 250–400 epochs, demonstrating effective early stopping.



(a) Training and Validation Loss

(b) Validation F1 Across Epochs

Figure 3: Training Dynamics Across Epochs

Threshold sweeping was applied to optimise F1 performance. This improved the Active F1 score by a few percentage points compared to the default threshold of 0.5.

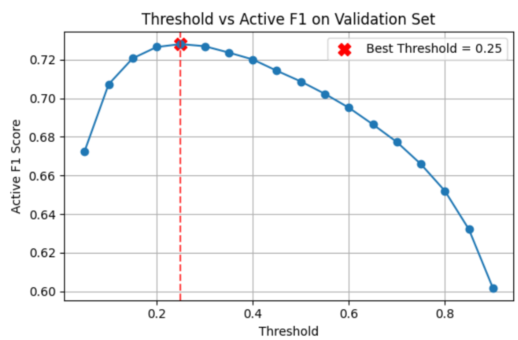


Figure 4: Threshold vs F1 Curve

4.2 Error Analysis

Figure 5 shows that the Transformer model with CQT input performs well across many mid-range notes, with F1 scores frequently exceeding 0.8. However, performance drops for both low and high extremes, with high notes (above middle octave) suffering disproportionately despite similar class imbalance. This suggests that challenges beyond data frequency—such as acoustic ambiguity or shorter durations—may hinder detection of high-pitched notes.

Manual inspection of misclassified frames revealed two recurring errors: **onset blurring**, where short notes in dense passages were missed; and **missed rare notes**, often ignored due to few training examples. These highlight the roles of class imbalance and temporal complexity in model limitations

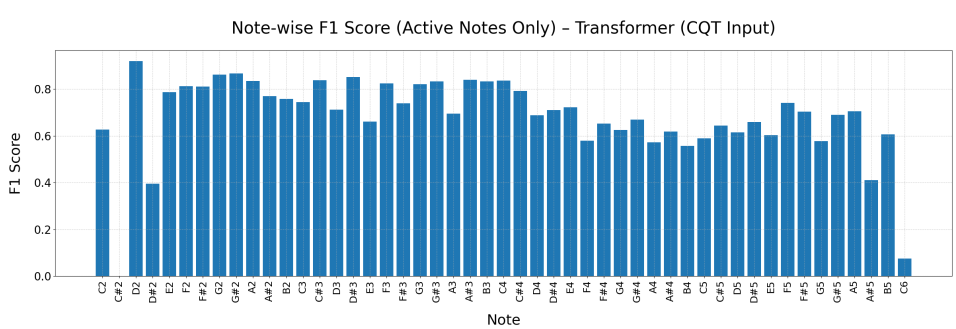


Figure 5: Per-note F1 scores using Transformer (CQT). High accuracy for low to mid-range

5 Conclusion, Achievements, and Future Work

This project tackled frame-wise piano note classification using lightweight deep learning models applied to polyphonic audio. The core aim was to evaluate the effectiveness of GRU and Transformer architectures across two input types: Constant-Q Transform (CQT) and Wav2Vec2 embeddings.

The main achievements include:

- Full implementation of all models from scratch—CNN baseline, GRU, and Transformer—without reliance on pre-trained components.
- Demonstration that lightweight Transformers trained on CQT outperformed GRUs and CNNs, achieving the highest Active F1 score of 0.68.
- Validation that pitch-aligned CQT inputs are significantly more effective for AMT than Wav2Vec2 embeddings, which struggled due to lower temporal resolution and speech-focused pretraining.
- Analysis of common failure modes, including onset blurring and poor recall of rare or high-pitched notes.

These findings reinforce that CQT representations combined with attention-based architectures form a strong baseline for efficient and accurate AMT, particularly under computational constraints. A notable insight was the surprising underperformance of Wav2Vec2 in this domain, underscoring the importance of domain-specific feature alignment.

Another unexpected result was that the GRU did not outperform the CNN baseline, despite its capacity for temporal modeling. This may reflect the limited model depth or the challenge of capturing long-term dependencies with relatively small recurrent architectures—highlighting that even sequential tasks may not benefit from recurrence alone without sufficient complexity or context.

Future work could explore:

- Fine-tuning Wav2Vec2 embeddings on music datasets to enhance pitch and temporal alignment.
- Combining CQT with learned embeddings to capture complementary signal properties.
- Applying lightweight data augmentation or generating synthetic training data to mitigate class imbalance and improve rare note detection.
- Incorporating onset-informed models (e.g., Onsets and Frames) to improve detection in fast or dense musical passages.

In summary, this work provides a reproducible and extensible baseline for polyphonic AMT, demonstrating the value of interpretable and computationally efficient deep learning techniques in music transcription.

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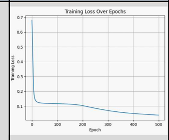
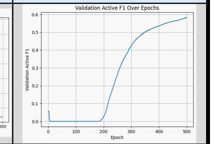
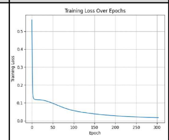
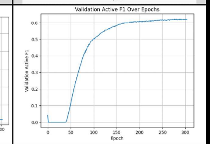
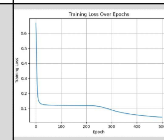
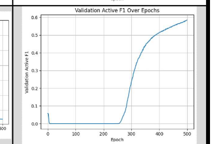
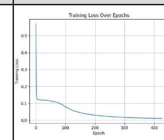
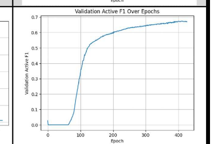
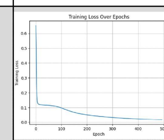
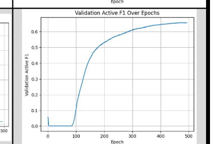
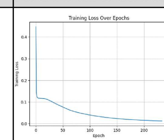
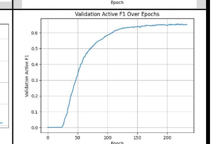
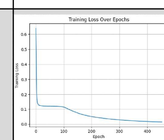
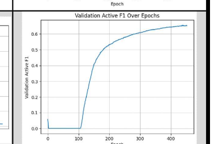
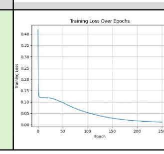
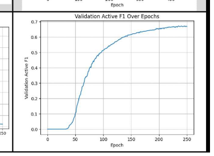
A Appendix A: Full Excel Table with Images

Input	Model	Hidden Size	Layers	Heads	Dropout	Learning Rate	Batch size	Threshold	Max Epoches	Early Stopping Patience	Best Epoch	Stopped Early	Best Val F1	Training Loss Over Epoches	Validation Active F1 Over Epoches
CQT	GRU	128	2	N/A	0.1	1.00E-04	1.28E+02	0.3	500	15	500	No	0.5828		
CQT	GRU	128	2	N/A	0.1	5.00E-04	1.28E+02	0.3	500	15	289	Yes	0.6246		
CQT	GRU	128	3	N/A	0.3	1.00E-04	1.28E+02	0.3	500	15	500	No	0.5846		
CQT	GRU	128	3	N/A	0.3	5.00E-04	1.28E+02	0.3	500	15	412	Yes	0.6752		
CQT	GRU	256	2	N/A	0.1	1.00E-04	1.28E+02	0.3	500	15	479	Yes	0.657		
CQT	GRU	256	2	N/A	0.1	5.00E-04	1.28E+02	0.3	500	15	219	Yes	0.6574		
CQT	GRU	256	3	N/A	0.3	1.00E-04	1.28E+02	0.3	500	15	438	Yes	0.6547		
CQT	GRU	256	3	N/A	0.3	5.00E-04	1.28E+02	0.3	500	15	236	Yes	0.6721		

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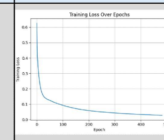
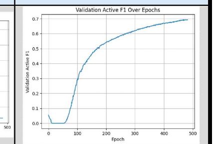
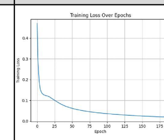
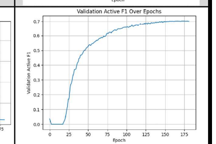
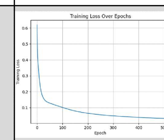
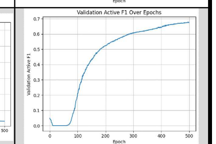
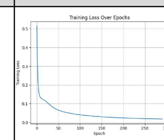
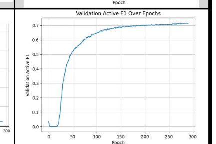
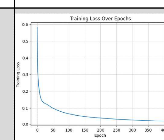
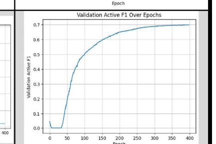
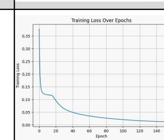
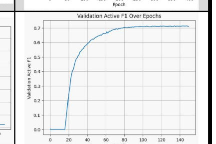
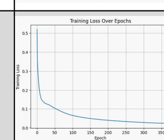
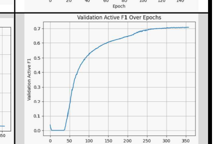
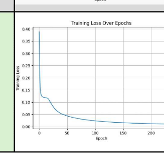
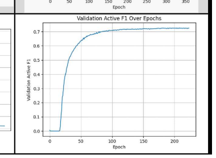
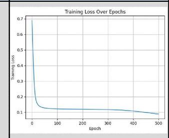
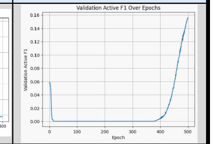
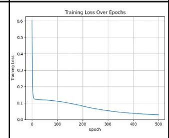
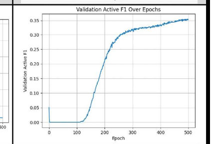
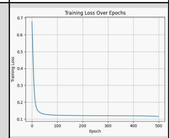
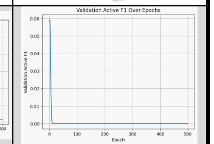
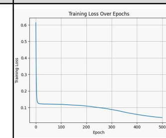
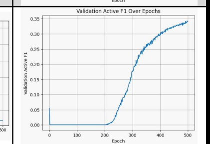
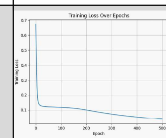
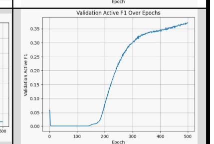
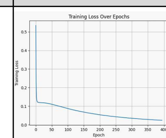
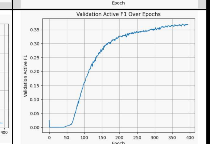
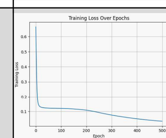
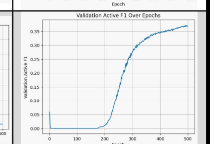
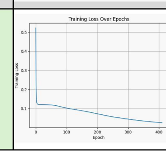
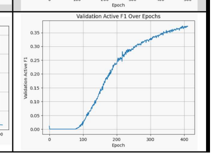
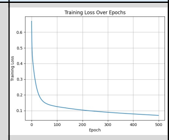
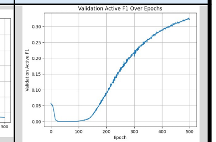
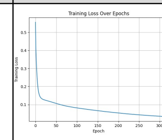
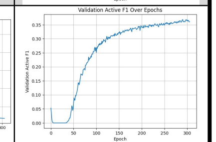
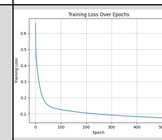
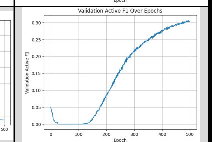
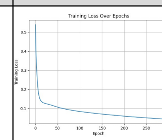
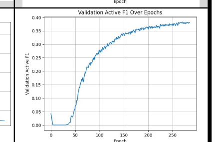
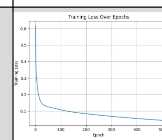
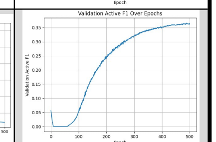
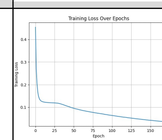
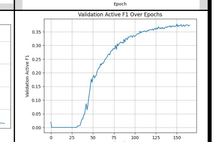
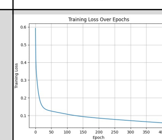
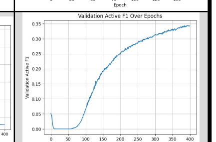
CQT	Transformer	128	2	2	0.1	1.00E-04	1.28E+02	0.3	500	15	468	Yes	0.6942		
CQT	Transformer	128	2	2	0.1	5.00E-04	1.28E+02	0.3	500	15	167	Yes	0.7018		
CQT	Transformer	128	3	4	0.3	1.00E-04	1.28E+02	0.3	500	15	499	No	0.6779		
CQT	Transformer	128	3	4	0.3	5.00E-04	1.28E+02	0.3	500	15	276	Yes	0.7165		
CQT	Transformer	256	2	2	0.1	1.00E-04	1.28E+02	0.3	500	15	384	Yes	0.7006		
CQT	Transformer	256	2	2	0.1	5.00E-04	1.28E+02	0.3	500	15	135	Yes	0.7146		
CQT	Transformer	256	3	4	0.3	1.00E-04	1.28E+02	0.3	500	15	343	Yes	0.7087		
CQT	Transformer	256	3	4	0.3	5.00E-04	1.28E+02	0.3	500	15	209	Yes	0.7265		

Input	Model	Hidden Size	Layers	Heads	Dropout	Learning Rate	Batch size	Threshold	Max Epoches	Early Stopping Patience	Best Epoch	Stopped Early	Best Val F1	Training Loss Over Epoches	Validation Active F1 Over Epoches
Wav2Vec2	GRU	128	2	N/A	0.1	1.00E-04	1.28E+02	0.3	500	15	500	No	0.1562		
Wav2Vec2	GRU	128	2	N/A	0.1	5.00E-04	1.28E+02	0.3	500	15	499	No	0.3551		
Wav2Vec2	GRU	128	3	N/A	0.3	1.00E-04	1.28E+02	0.3	500	15	500	No	0		
Wav2Vec2	GRU	128	3	N/A	0.3	5.00E-04	1.28E+02	0.3	500	15	498	No	0.3429		
Wav2Vec2	GRU	256	2	N/A	0.1	1.00E-04	1.28E+02	0.3	500	15	500	No	0.3719		
Wav2Vec2	GRU	256	2	N/A	0.1	5.00E-04	1.28E+02	0.3	500	15	380	Yes	0.3707		
Wav2Vec2	GRU	256	3	N/A	0.3	1.00E-04	1.28E+02	0.3	500	15	493	No	0.372		
Wav2Vec2	GRU	256	3	N/A	0.3	5.00E-04	1.28E+02	0.3	500	15	396	Yes	0.3742		

Input	Model	Hidden Size	Layers	Heads	Dropout	Learning Rate	Batch size	Threshold	Max Epoches	Early Stopping Patience	Best Epoch	Stopped Early	Best Val F1	Training Loss Over Epoches	Validation Active F1 Over Epoches
Wav2Vec2	Transformer	128	2	2	0.1	1.00E-04	1.28E+02	0.3	500	15	496	No	0.3267		
Wav2Vec2	Transformer	128	2	2	0.1	5.00E-04	1.28E+02	0.3	500	15	291	Yes	0.3686		
Wav2Vec2	Transformer	128	3	4	0.3	1.00E-04	1.28E+02	0.3	500	15	495	No	0.3066		
Wav2Vec2	Transformer	128	3	4	0.3	5.00E-04	1.28E+02	0.3	500	15	271	Yes	0.3828		
Wav2Vec2	Transformer	256	2	2	0.1	1.00E-04	1.28E+02	0.3	500	15	490	No	0.3662		
Wav2Vec2	Transformer	256	2	2	0.1	5.00E-04	1.28E+02	0.3	500	15	151	Yes	0.3782		
Wav2Vec2	Transformer	256	3	4	0.3	1.00E-04	1.28E+02	0.3	500	15	385	Yes	0.344		
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